Faculty of Engineering & Technology Electrical & Computer Engineering Department

ENEE4113 - COMMUNICATION LAB

Report#1

Experiment 2 : Double-side and Single-side Band Modulation

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Abstract

The purpose of this experiment was to facilitate the practical application of theoretical knowledge of Double-side Band (DSB) and Single-side Band (SSB) modulation methods in communication systems. In order to achieve this, the experiment aimed to develop a better understanding of the fundamentals of communication and the practical steps required to create DSB suppressed carrier (DSBsc) modulation. Additionally, the experiment also encompassed a study of the different types of SSB modulation, including SSB suppressed carrier (SSBsc) and SSB reduced carrier (SSBrc), and the practical methods used to create each type. The experiment also delved into the practical aspects of demodulating these methods and analyzed whether they were affected by non-coherent problems.

Contents

Table Of Figures

1.Theory

1.1 Double-side band modulation

Definition: DSB-SC is an amplitude modulated wave transmission scheme in which only sidebands are transmitted and the carrier is not transmitted as it gets suppressed. DSB-SC is an acronym for Double Sideband Suppressed Carrier.
The carrier does not contain any information and its transmission results in loss of power. Only sidebands are transmitted that contains information. This results in saving of power used in transmission.
This saved power can be inserted into the 2 sidebands. Hence, ensuring a stronger signal that transmits over long distances. As during suppression, the baseband signal does not get affected in any way.

As we know that transmission power and bandwidth are the two important parameters in a communication system. Thus, in order to save power and bandwidth, DSB-SC modulation technique is adopted.

1.1.1 Generation of DSB-SC signal

Let’s have a look at the block diagram of the DSB-SC system shown below:

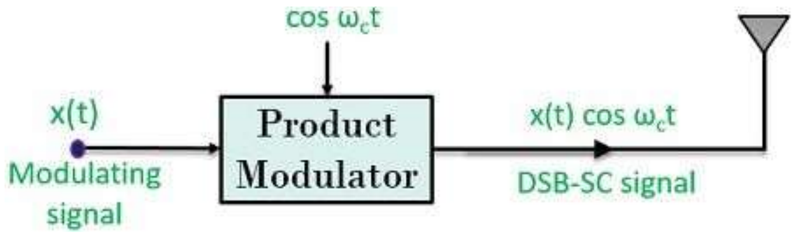

Figure 1:Generation of DSB-SC signal

Here, by observing the above figure, we can say that a product modulator generates a DSB-SC signal.

The signal is obtained by the multiplication of baseband signal x(t) with carrier signal cos ω_ct

By frequency shifting property of Fourier transform-

$$x(t) \cos \omega_c t \longleftrightarrow \frac{1}{2}[\, X\,(\omega + \omega_c) + (\omega - \omega_c)]$$

From the above equation, it is clear that only 2 components are present in the spectrum. These two are the two sidebands that are placed at **+ω_c** and **-ω_c**.

Let's have a look at the pictorial representation of a waveform for DSB-SC system-

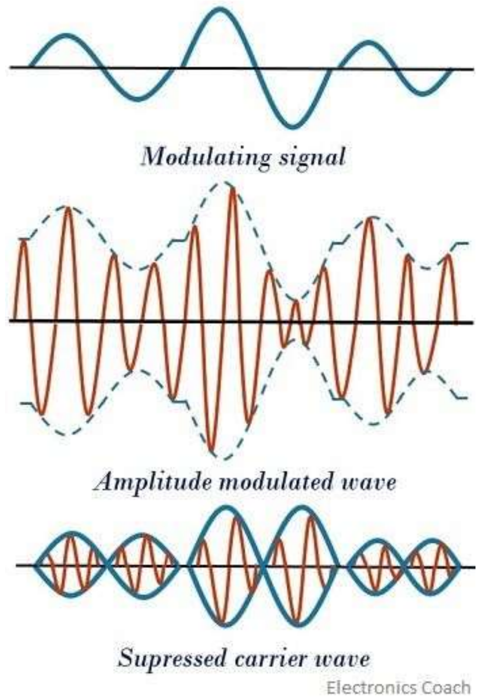

Figure 2: DSB-SC system

1.1.2 DSB-SC Demodulation:

We use coherent detection for demodulation. Basically we multiply our received signal with a signal identical to the carrier (preferably with synchronized frequency and phase) .

$$v(t) = u(t) * A_c' cos(2\pi f_c t + \Theta)$$
$$v(t) = A_c m(t) cos(2\pi f_c t) * A_c' cos(2\pi f_c t + \Theta)$$
$$v(t) = \frac{A_c A_c' m(t)}{2}[cos(4\pi f_c t + \Theta) + cos(\Theta)]$$
$$v(t) = \frac{A_c A_c' m(t)}{2} cos(4\pi f_c t + \Theta) + \frac{A_c A_c' m(t)}{2} cos(\Theta)$$

The second term is proportional to m(t) and therefore has lower frequency (f_m). The first term has a frequency of $2f_c$. Using a low pass filter on v(t) can extract the message signal successfully.

$$V(f) = \frac{A_c A_c'}{4}[e^{i\Theta} M(f - 2f_c) + e^{-i\Theta} M(f + 2f_c)] + \frac{A_c A_c' M(f)}{4} cos(\Theta)$$

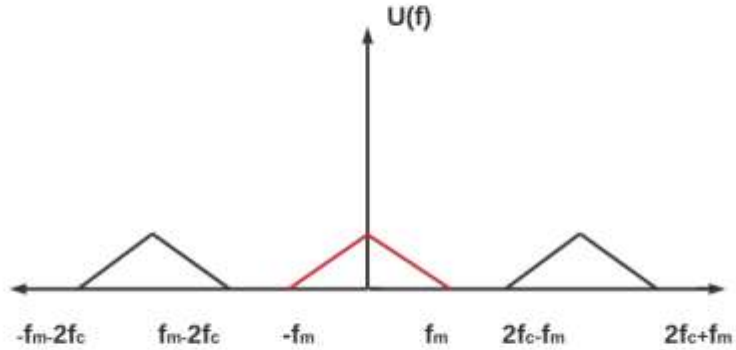

Figure 3: DSB-SC demodulation

Note: The magnitudes are not true to value. It is just a representation for better understanding. The frequency highlighted in red is our desired message signal. We need to eliminate the others. Looking at the above diagram, we can say the cut off frequency of the lowpass filter must be at least f_m (or W so that the message signal can pass) and it must be less than $2f_c$-f_m(or $2f_c$-W).

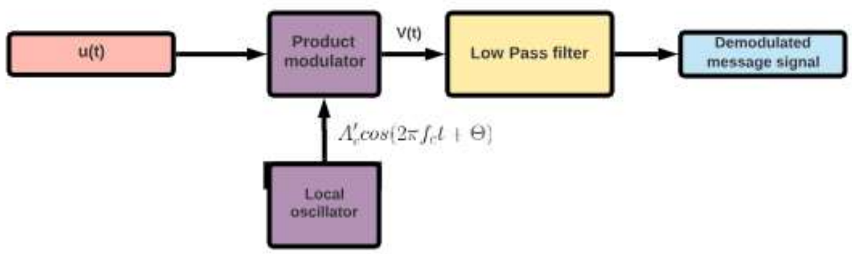

Figure 4: DSB-SC demodulation system

1.2 Single-side Band Modulation

A Sideband is a band of frequencies, containing power, which are the lower and higher frequencies of the carrier frequency. Both the sidebands contain the same information. The representation of amplitude modulated wave in the frequency domain is as shown in the following figure.

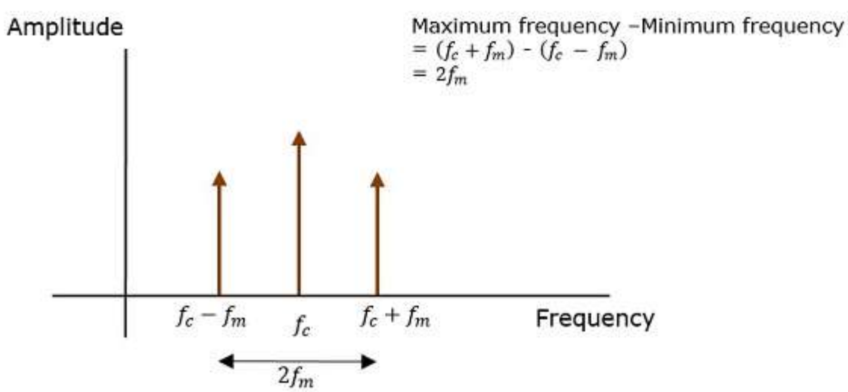

Figure 5:Amplitude modulated wave in the frequency domain

Both the sidebands in the image contain the same information. The transmission of such a signal which contains a carrier along with two sidebands, can be termed as Double Sideband Full Carrier system, or simply DSB-FC. It is plotted as shown in the following figure.

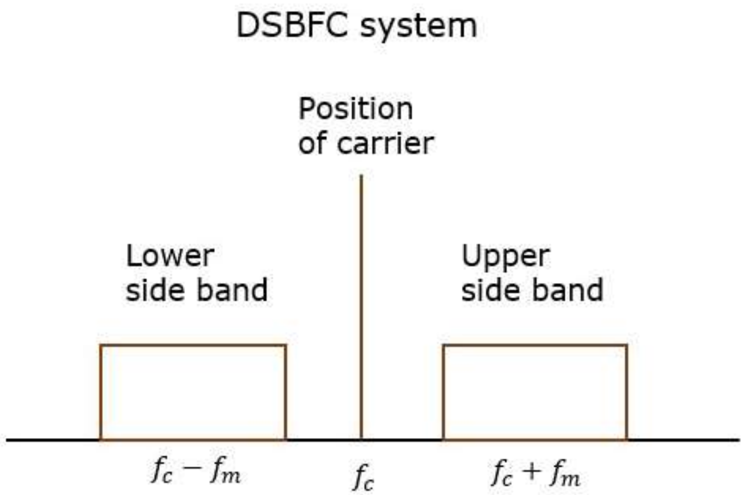

Figure 6: DSB-FC system

However, such a transmission is inefficient. Two-thirds of the power is being wasted in the carrier, which carries no information.

If this carrier is suppressed and the power saved is distributed to the two sidebands, such a process is called as Double Sideband Suppressed Carrier system, or simply DSBSC. It is plotted as shown in the following figure.

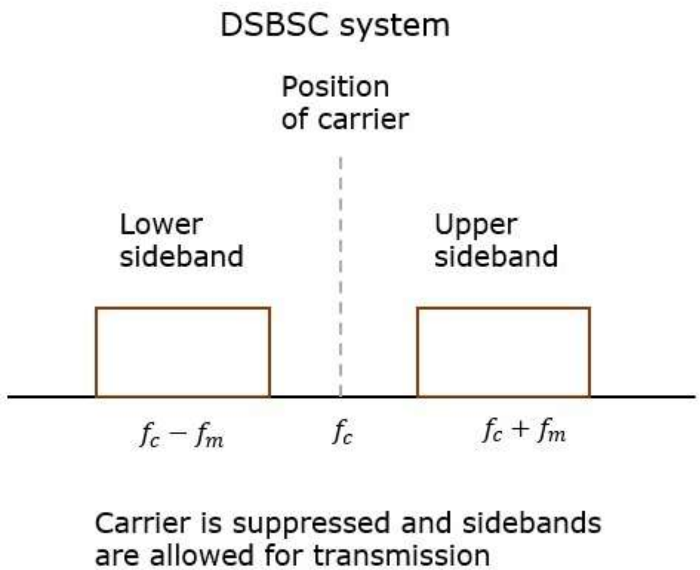

Figure 7: DSB-SC system

Now, we get an idea that, as the two sidebands carry the same information twice, why can't we suppress one sideband. Yes, this is possible.

The process of suppressing one of the sidebands, along with the carrier and transmitting a single sideband is called as Single Sideband Suppressed Carrier system, or simply SSB-SC or SSB. It is plotted as shown in the following figure.

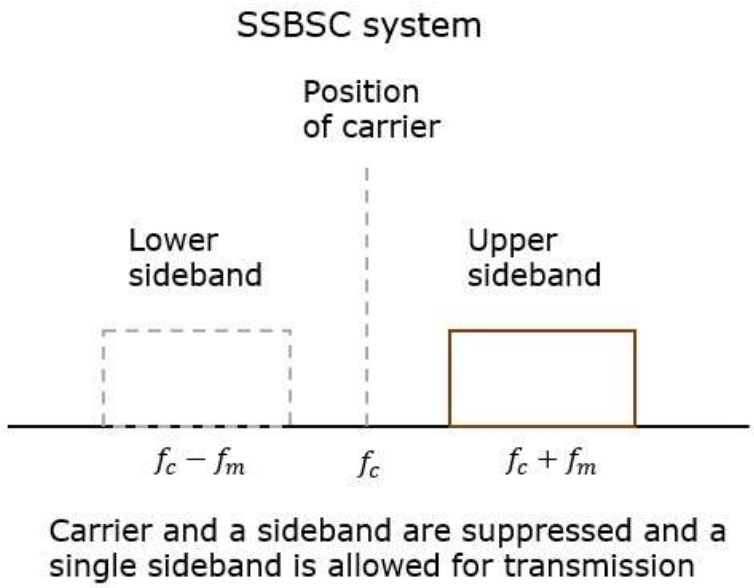

Figure 8:SSB-SC system

This SSB-SC or SSB system, which transmits a single sideband has high power, as the power allotted for both the carrier and the other sideband is utilized in transmitting this Single Sideband (SSB).

Hence, the modulation done using this SSB technique is called as SSB Modulation

1.2.1 Mathematical Expressions

Let us consider the same mathematical expressions for the modulating and the carrier signals as we have considered in the earlier chapters.

i.e., Modulating signal

$$m(t) = A_m \cos(2\pi f_m t)$$

Carrier signal

$$c(t) = A_c \cos(2\pi f_c t)$$

Mathematically, we can represent the equation of SSBSC wave as

$$s(t) = \frac{A_m A_c}{2} \cos[2\pi (f_c + f_m) t]$$ for the upper sideband

Or

$$s(t) = \frac{A_m A_c}{2} \cos[2\pi (f_c - f_m) t]$$ for the lower sideband

1.2.2 SSB-SC Demodulation

The baseband or modulating signal can be recovered from the SSB-SC signal by using the synchronous detection. Coherent Demodulation of SSB signals ΦSSB (t) is multiplied with cos(ωc t) and passed through low pass filter to get back the original signal.

A true SSB demodulator must have the ability to select sidebands. All the methods of SSB generation so far discussed have their counterparts as demodulators. In this experiment you will be examining the phasing-type demodulator, shown in Fig.1.2.2 .

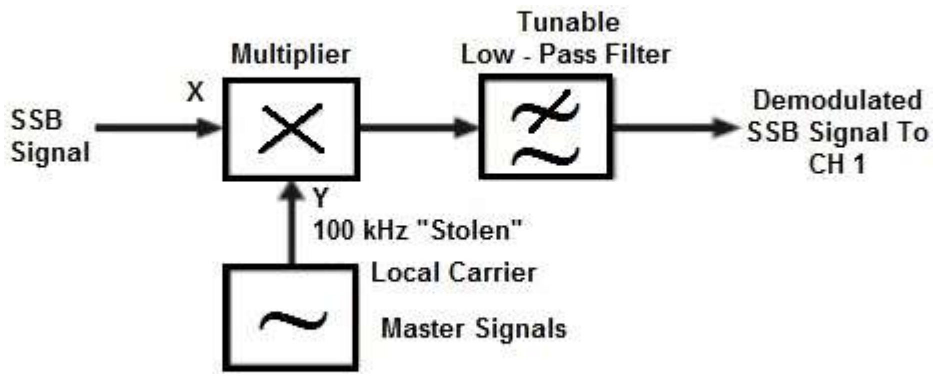

Figure 9 : SSB-SC demodulation

2.Procedure & Discussion

2.1 Double-side band suppressed carrier amplitude Modulation

2.1.1 Modulation (Time and Frequency domains)

We have assembled the components and use as shown in the figure below:

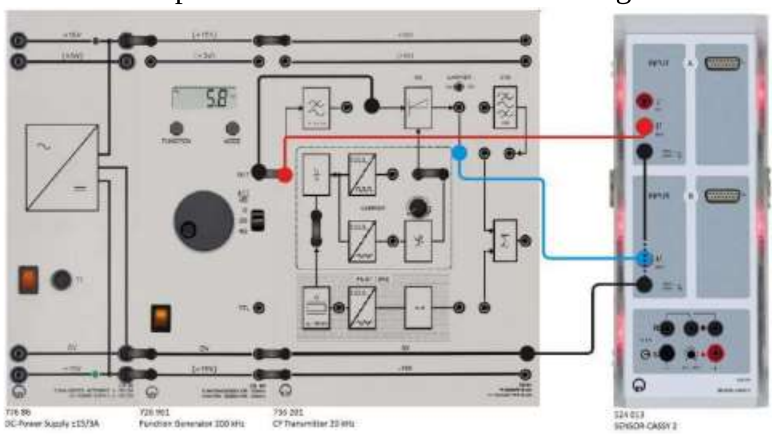

Figure 10: DSB modulation components

The message signal of the DSB a sinusoidal message VSS = 4V and fm = 2 kHz, after drawing the output of the modulated signal s(t) and the message signal , we get the following :

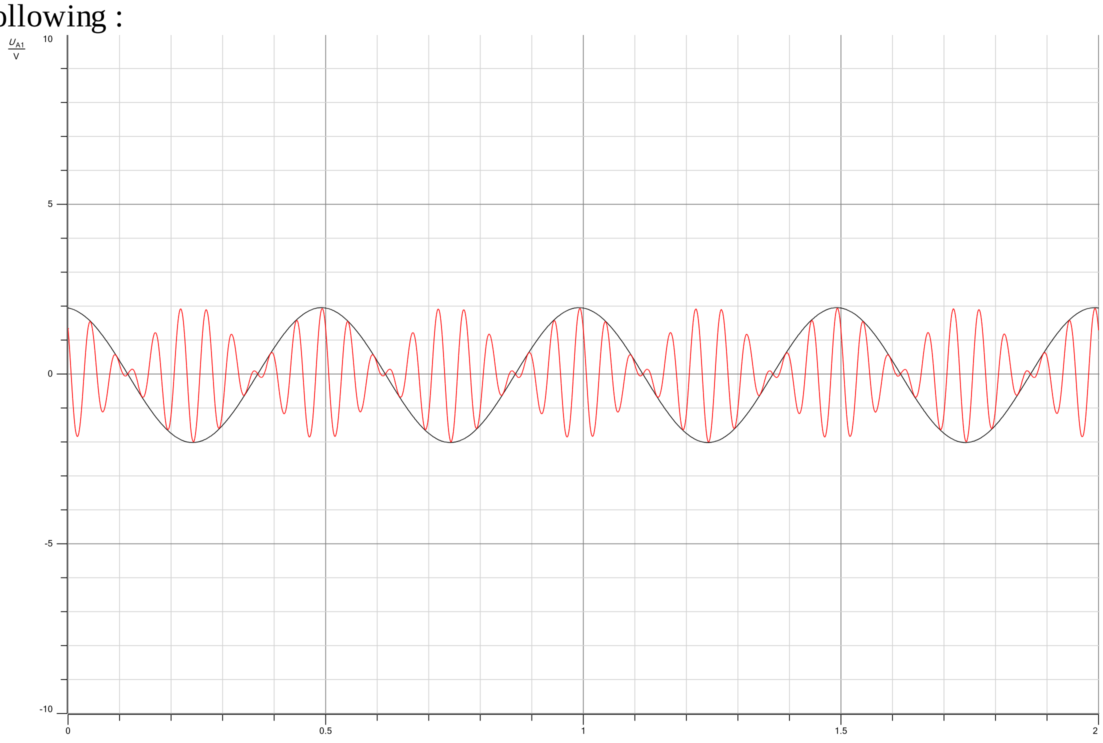

Figure 11:DSB output at 4v & 2kHz

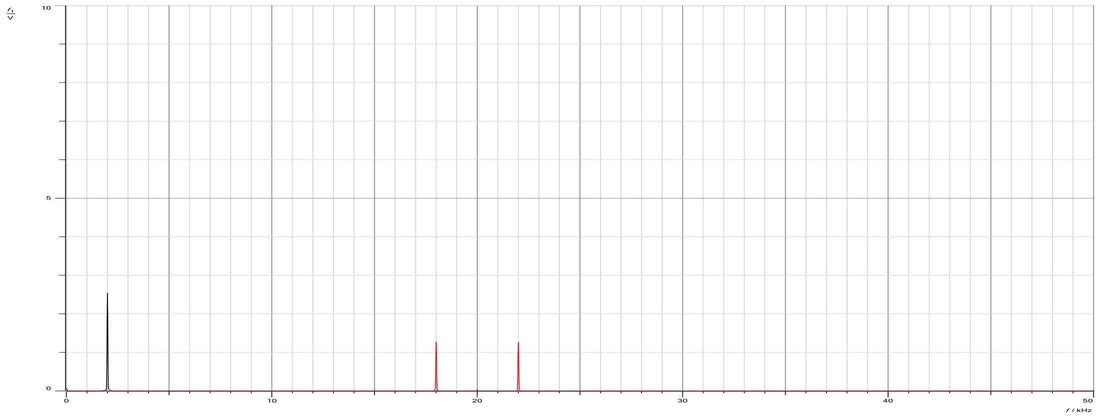

Figure 12: DSBsc output in frequency

The black signal in the figures represents the message signal m(t) in time and frequency domain. The red one is the modulated signal s(t), the time domain of s(t) indicates the signal that comes from multiplying the message with the carrier, which is s(t), and the bandwidth of s(t) from the frequency domain is 2000 Hz, which is 2 * fm. Also, we can see from the frequency domain of s(t) that the carrier is not transferred, resulting in a power efficiency of 100%. Furthermore, the modulated signal s(t) is focused on fc.

As for the effect of varying the frequency (fm) and the amplitude (Am) of the message signal:

- The effect of changing the frequency: kept Vss = 4V and change the message frequency to
  - ➢ 1kHz, and the following resulted:

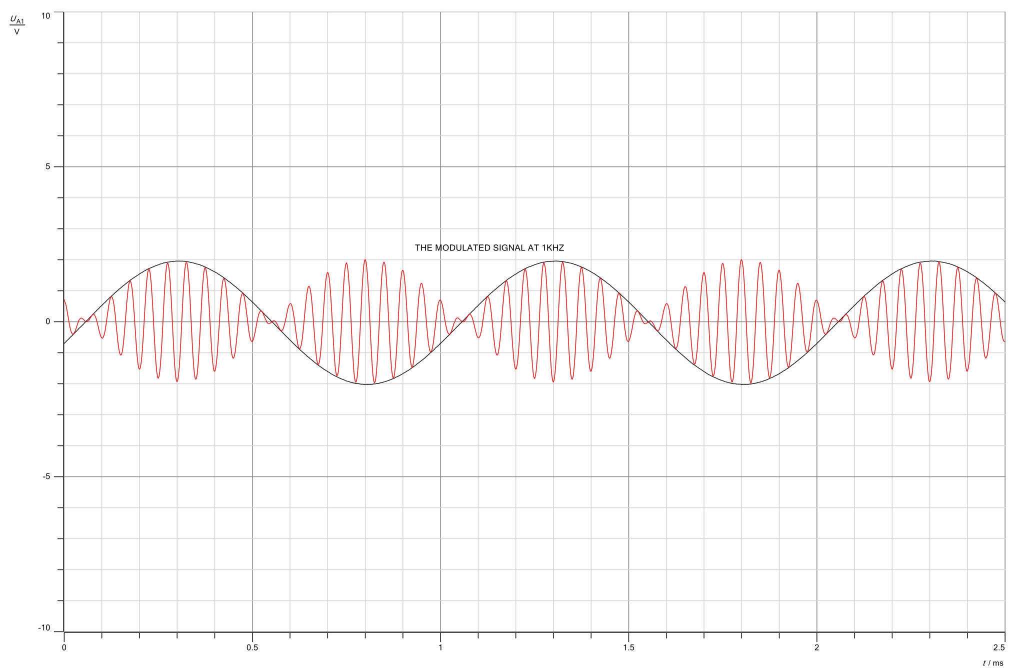

Figure 13: DSB output at 4v & 1kHz

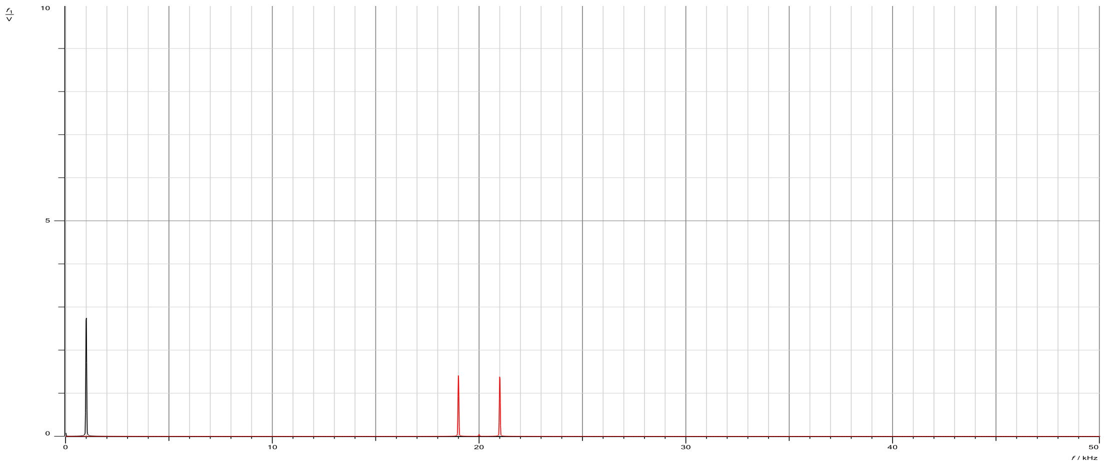

Figure 14: DSB output at 4v & 1kHz in freq. domain

- 3kHz, and the following resulted:

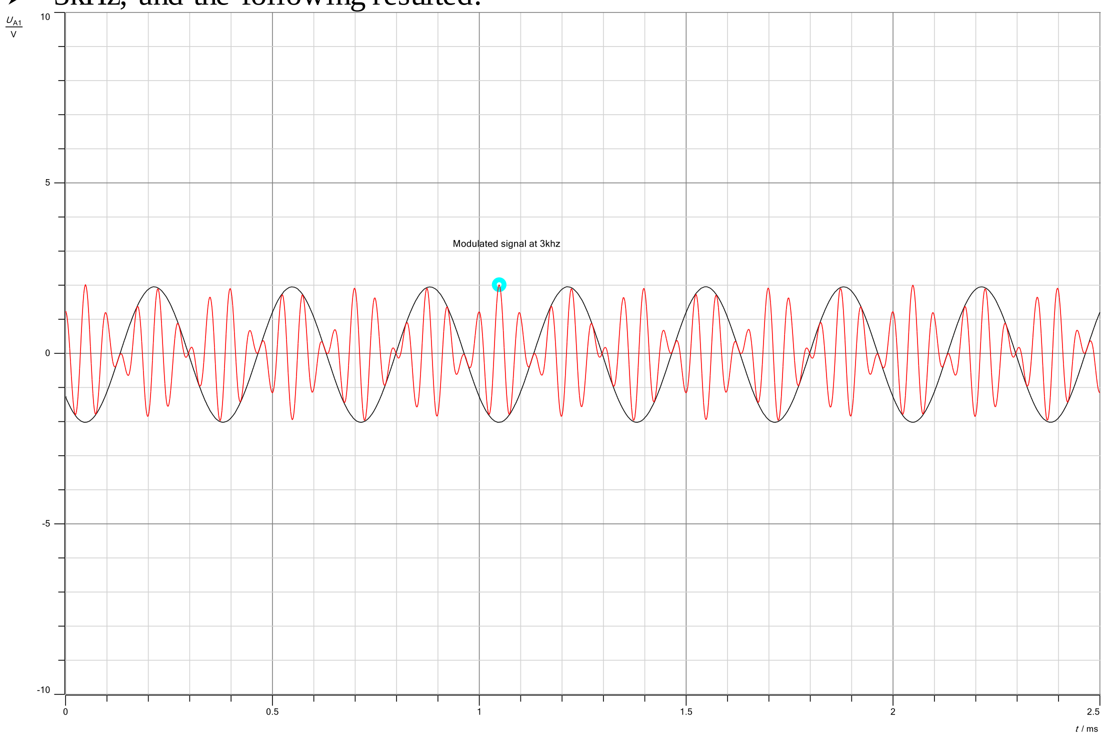

Figure 15: DSB output at 4v & 3kHz

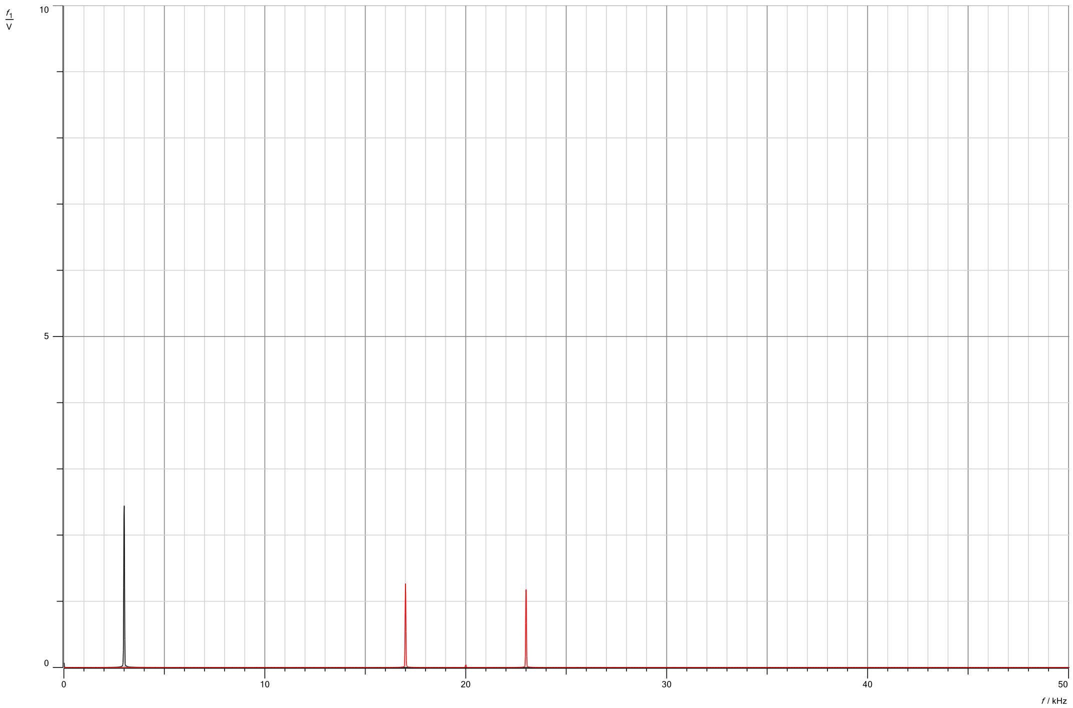

Figure 16: DSB output at 4v & 3kHz in freq. domain

As shown in the figures, changing fm only affected the bandwidth of the message m(t) as well as the bandwidth of s(t). Frequency changes the number of waves with respect to time, and the modulated signal waveform was shifted.

The effect of changing the amplitude: Kept fm = 2kHz and change the message Vss to

- 2V, and the following resulted:

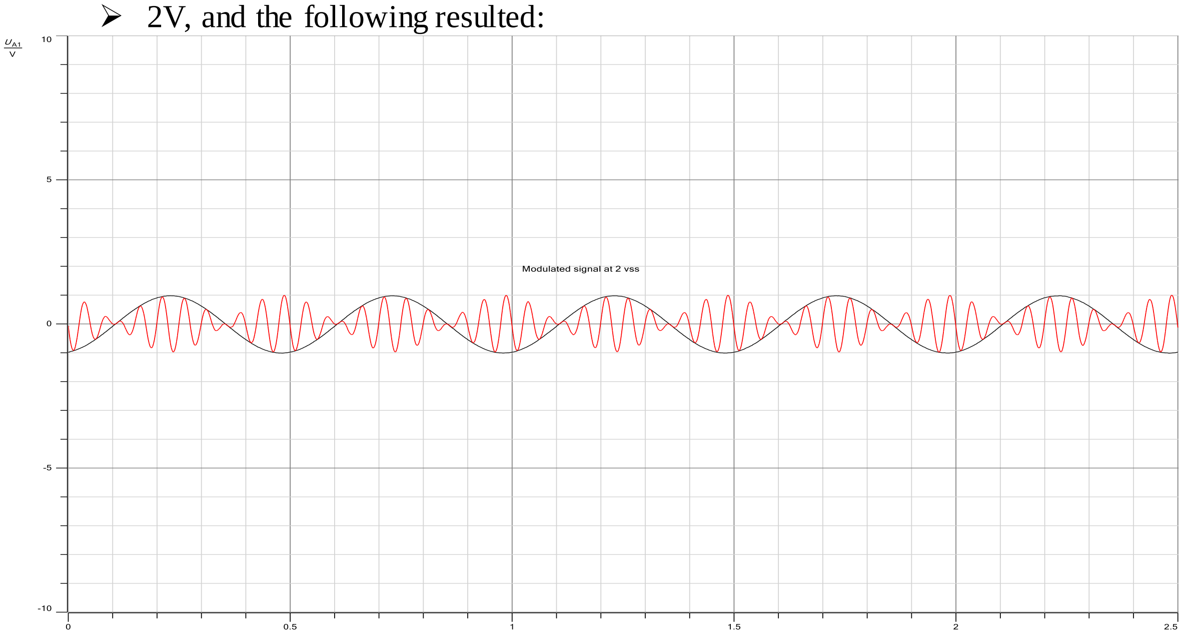

Figure 17: DSB output at 2v & 2kHz

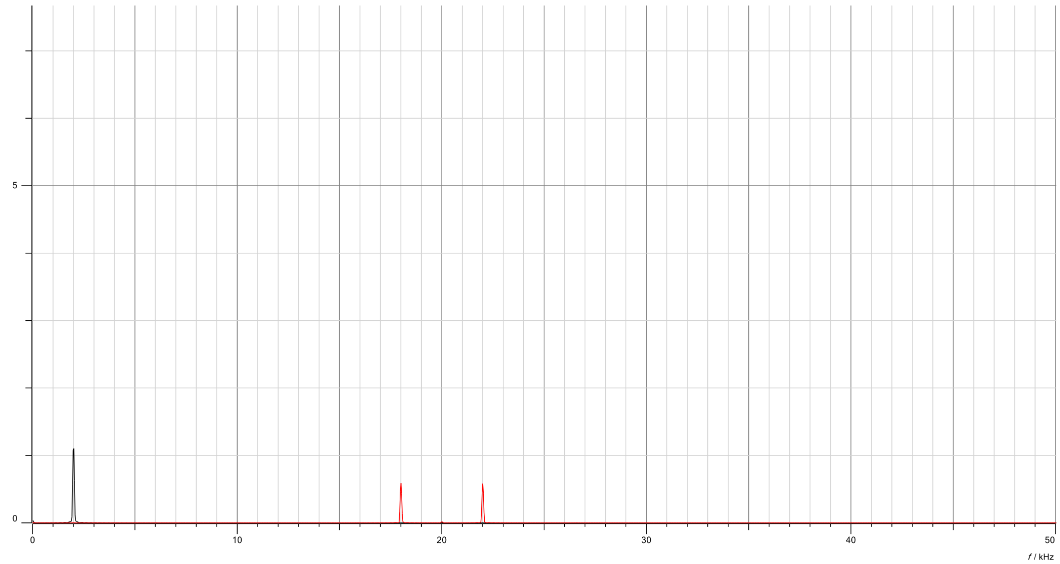

Figure 18: DSB output at 2v & 2kHz in freq. domain

- 6V, and the following resulted:

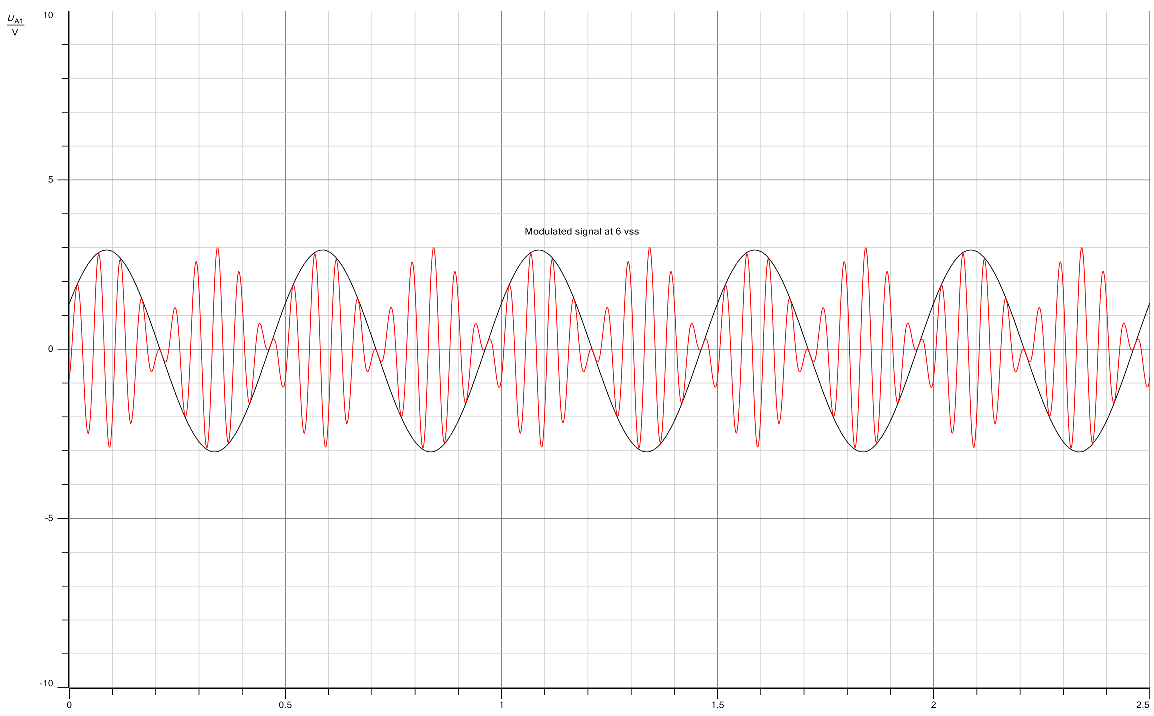

Figure 19 : DSB output at 6v & 2kHz

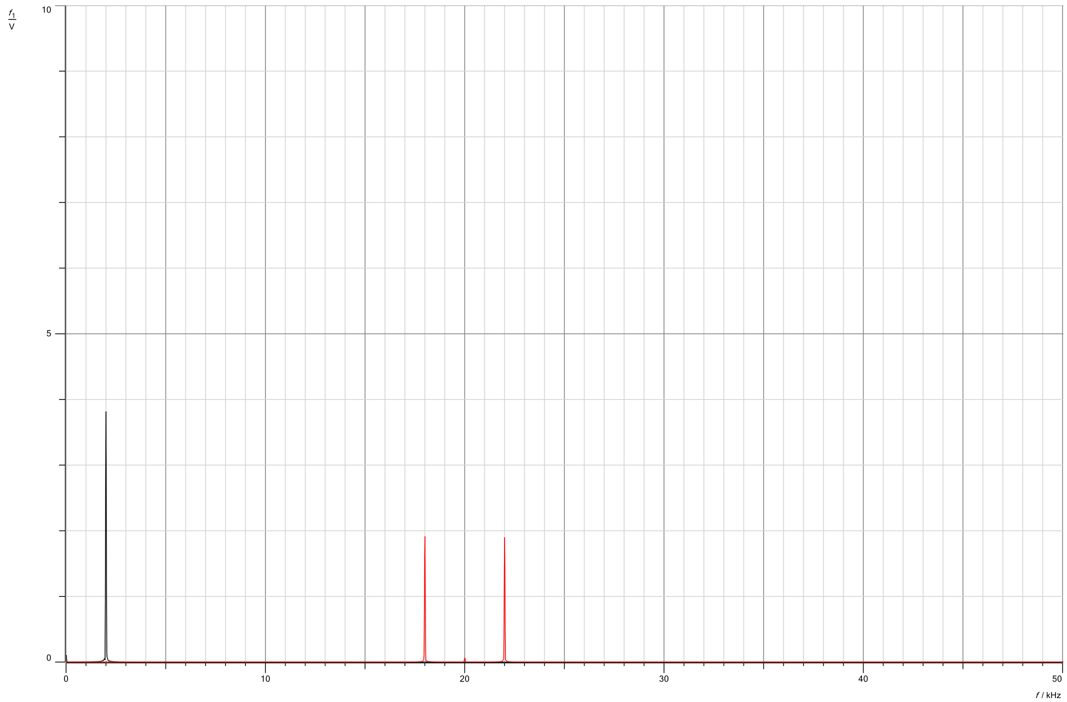

Figure 20: DSB output at 6v & 2kHz in freq. domain

As shown in the figure, changing Am only affected the amplitude of the message m(t) and the amplitude of s(t).

2.1.2 Double side band demodulation

2.1.2.1 Coherent Demodulation

We have assembled the components and use as shown in the figure below:

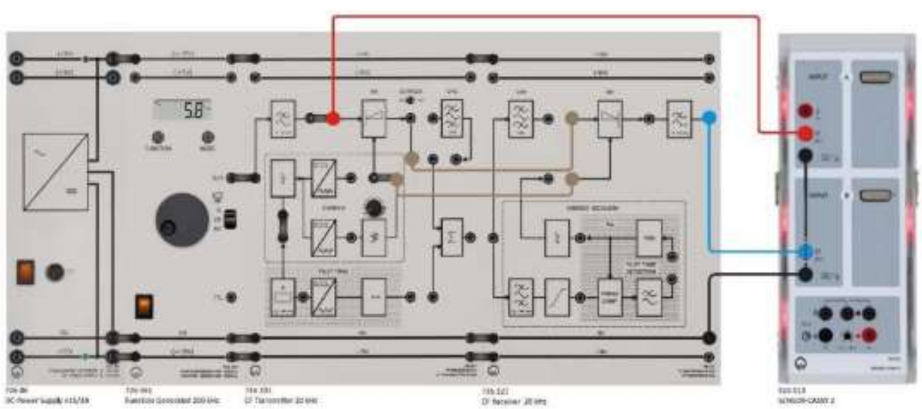

Figure 21: : DSB demodulation components

VSS = 4V and fm = 2 kHz , and Using a wire feed a sinusoidal carrier signal into the auxiliary carrier input of the demodulator to get the same frequency in the carrier signal for the demodulation.

And we got the following output:

- Before the output filter:

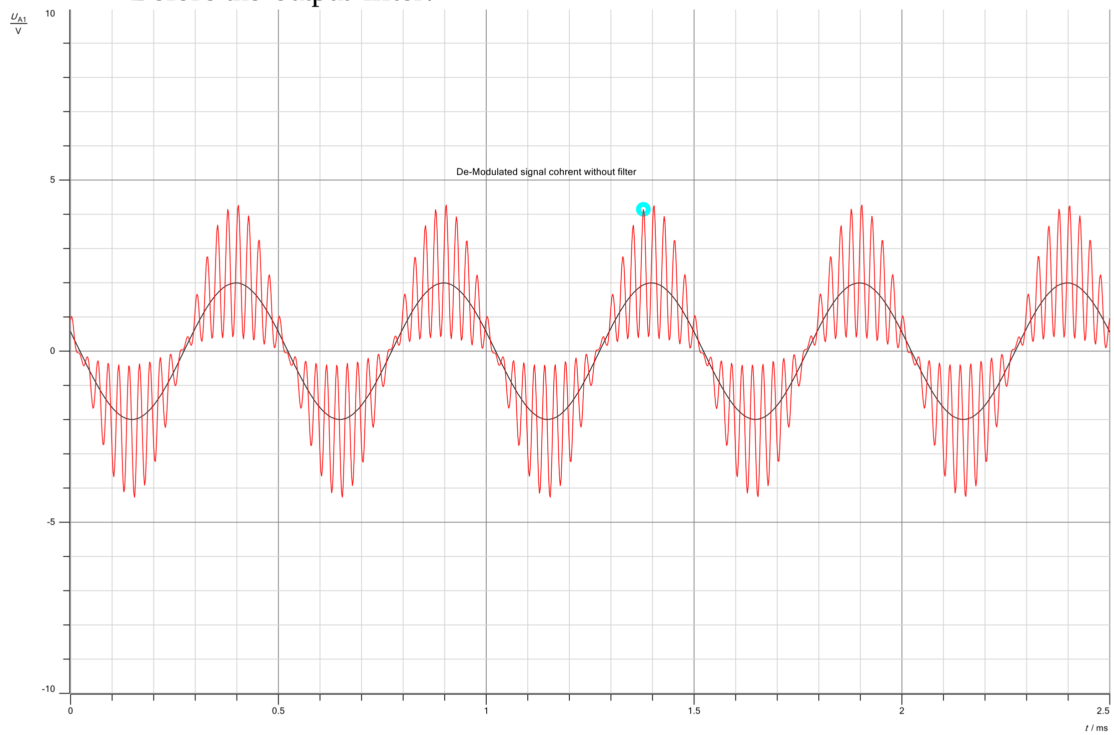

Figure 22: Demodulated signal without filter

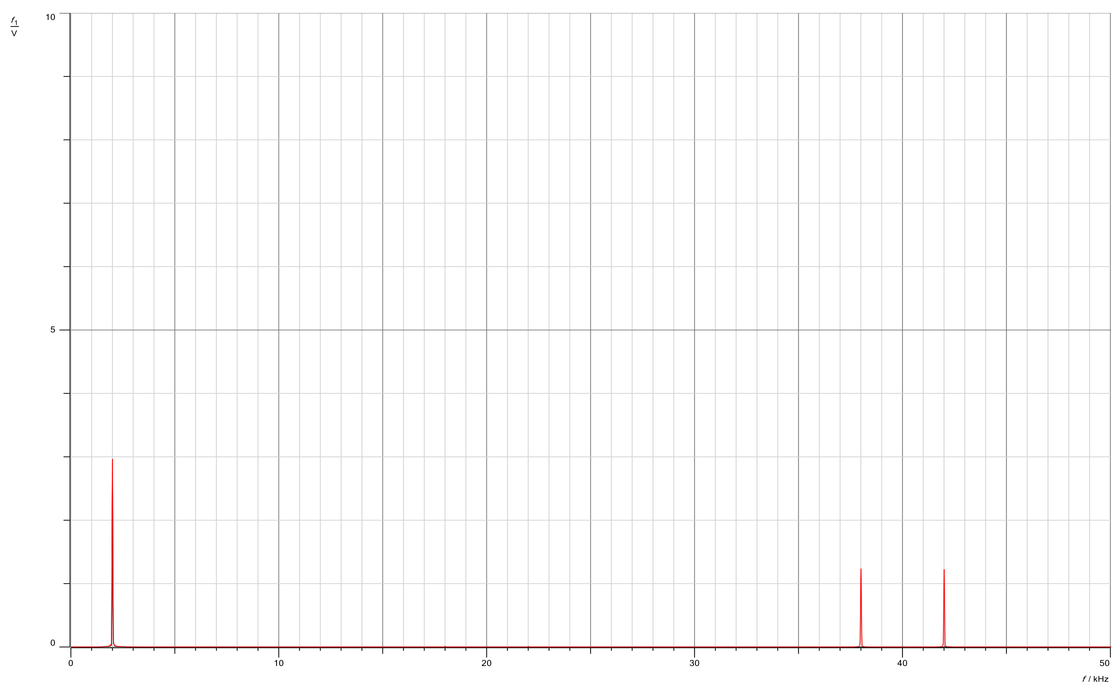

Figure 23: Demodulated signal without filter in freq. domain

➢ After the output filter:

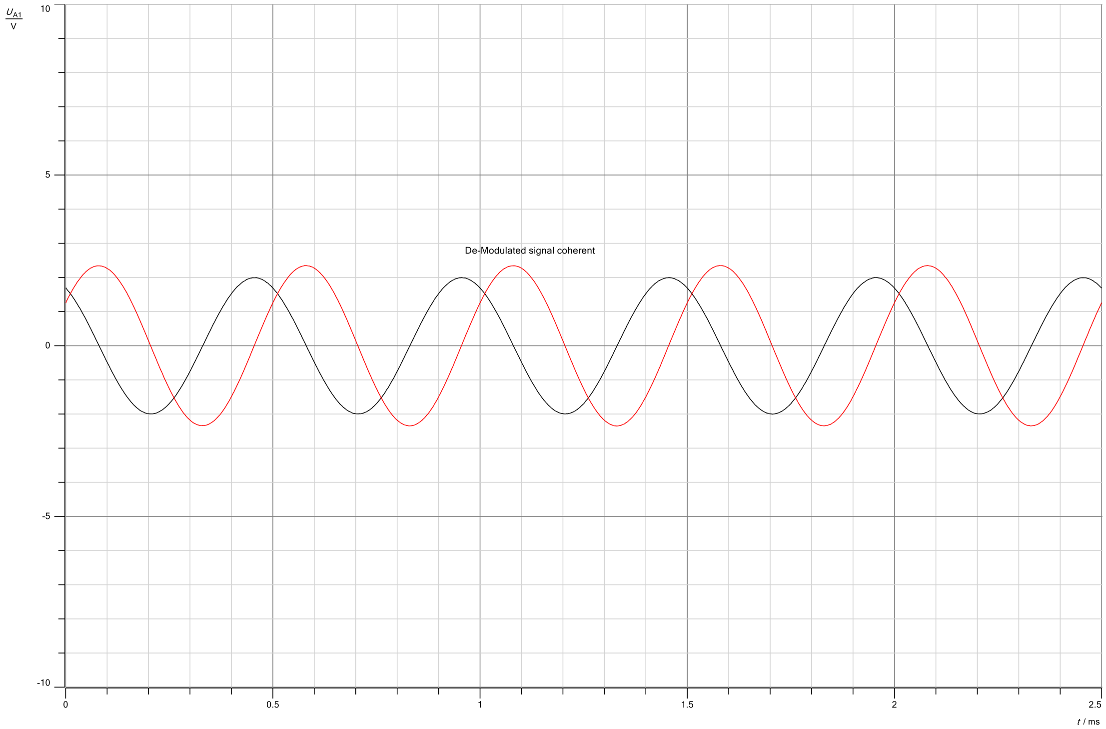

Figure 24: Demodulated signal with filter

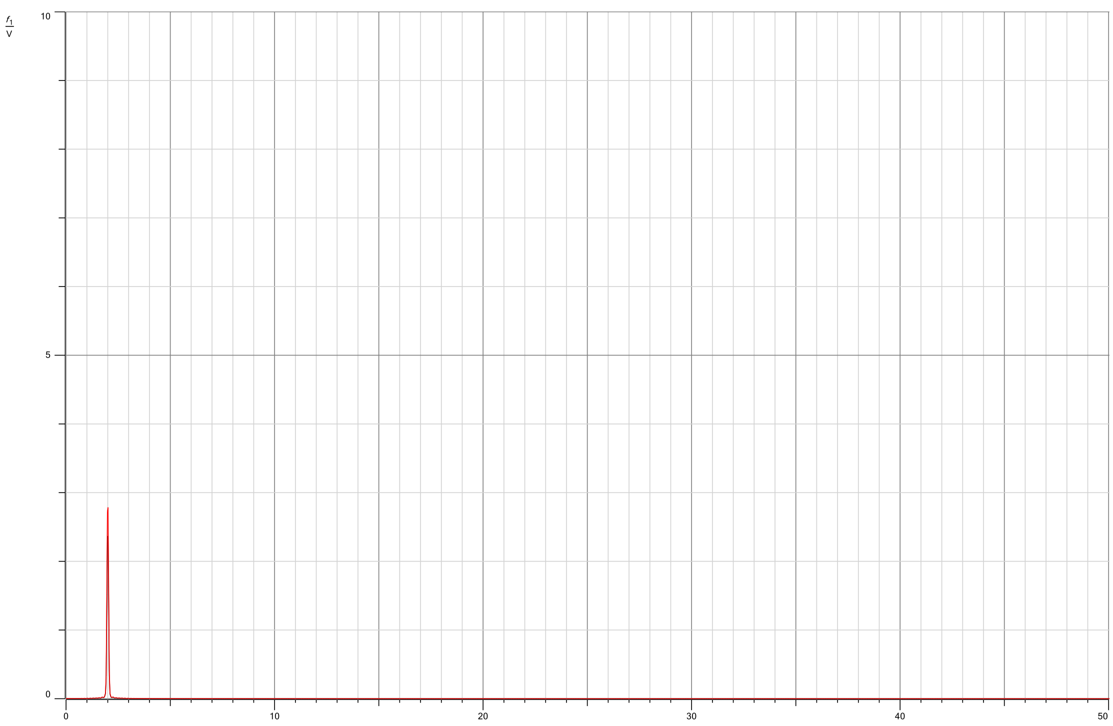

Figure 25: Demodulated signal with filter in freq. domain

Mathematically, the DSB-SC signal can be represented as:

$$s(t) = m(t)\cos(\omega_c t)$$

The demodulated signal can be represented as:

$$y(t) = \frac{1}{2}r(t) * h(t)$$

The frequency response of the low-pass filter is given by:

$$H(\omega) = \begin{cases} 1, & |\omega| < \omega_c \\ 0, & |\omega| > \omega_c \end{cases}$$

The Fourier transform of the output is:

$$Y(\omega) = \frac{1}{2}R(\omega)H(\omega)$$

We can observe from the frequency response of the low-pass filter that only frequency components within the message signal's bandwidth are passed through, while unwanted frequency components are rejected. As a result, for coherent demodulation of DSB signals, the receiver output filter is required.

2.1.2.2 Non-Coherent Demodulation

After made some simple changes in the meshes of the components to get the non-coherent and we got the following:

- Before the output filter:

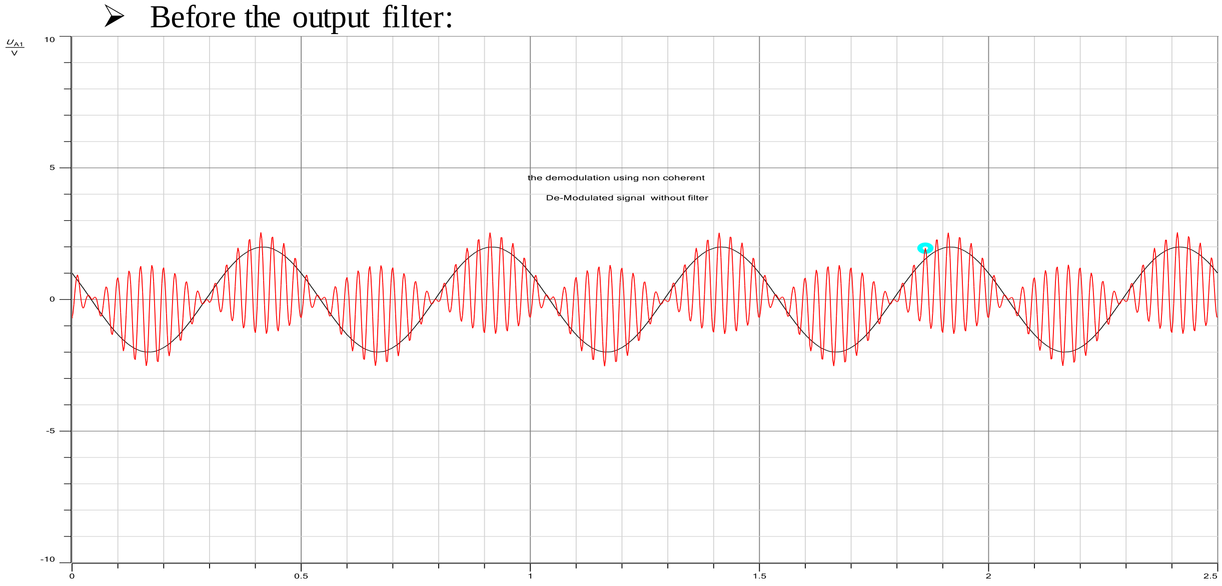

Figure 26: Non-Coherent Demodulation without filter

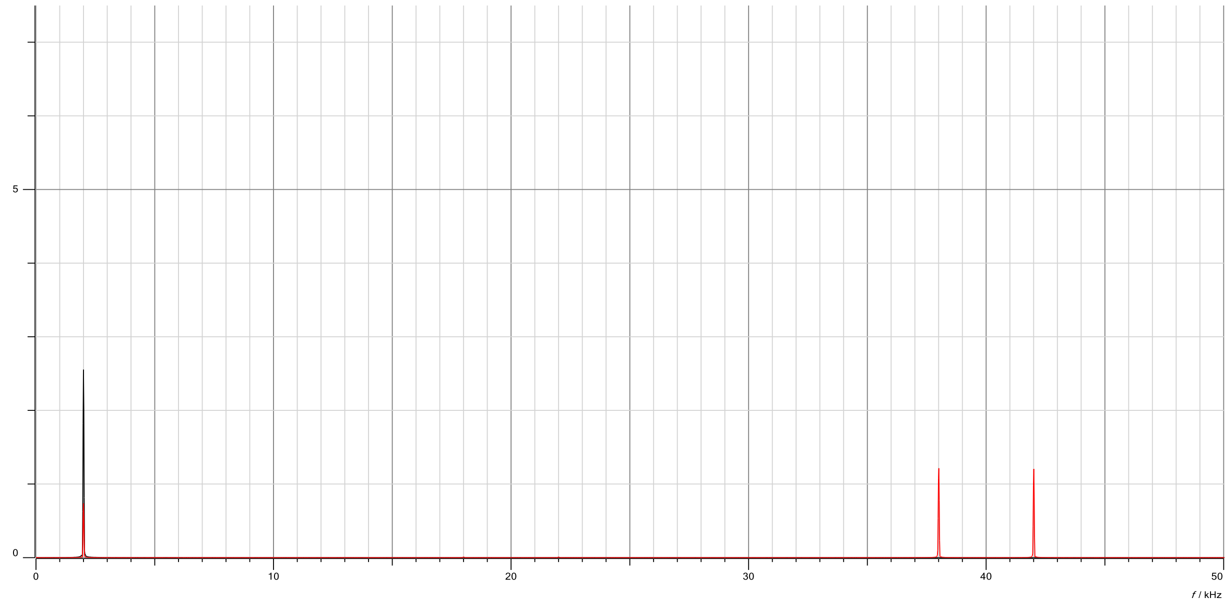

Figure 27: Non-Coherent Demodulation without filter in freq. domain

- After the output filter:

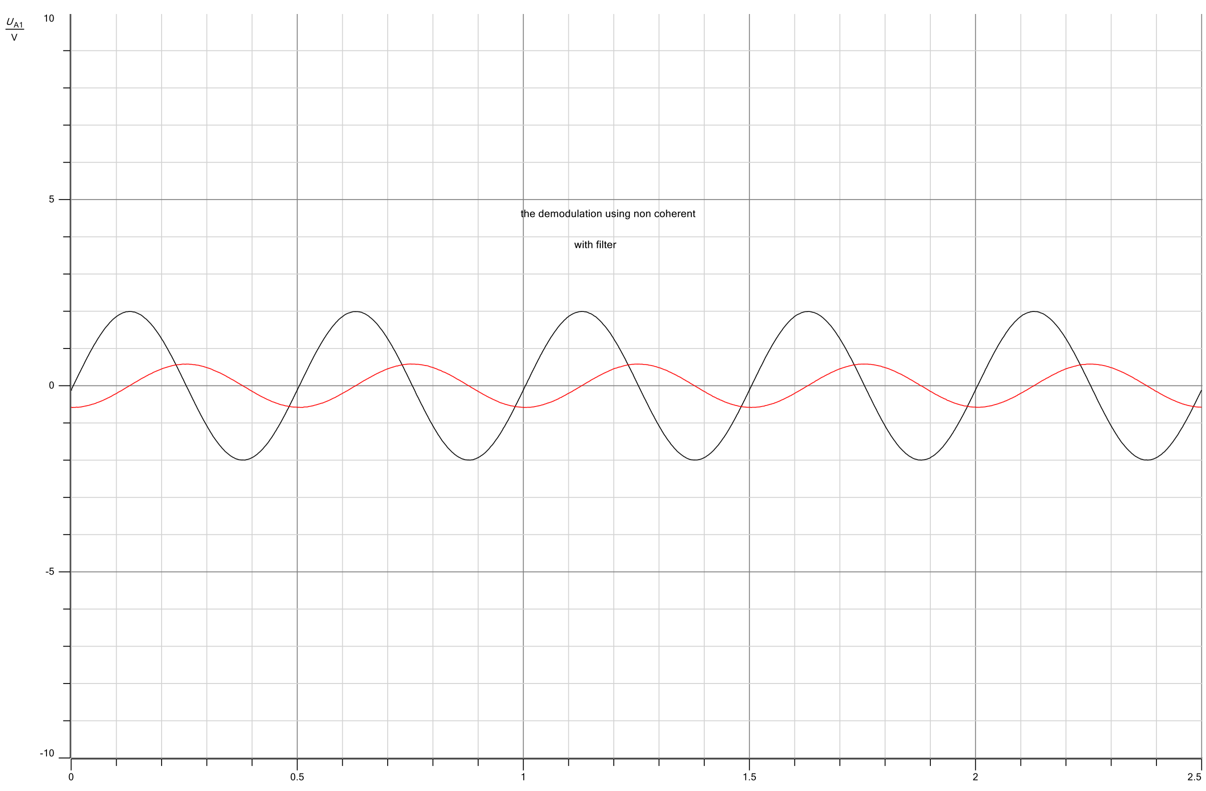

Figure 28: Non-Coherent Demodulation with filter

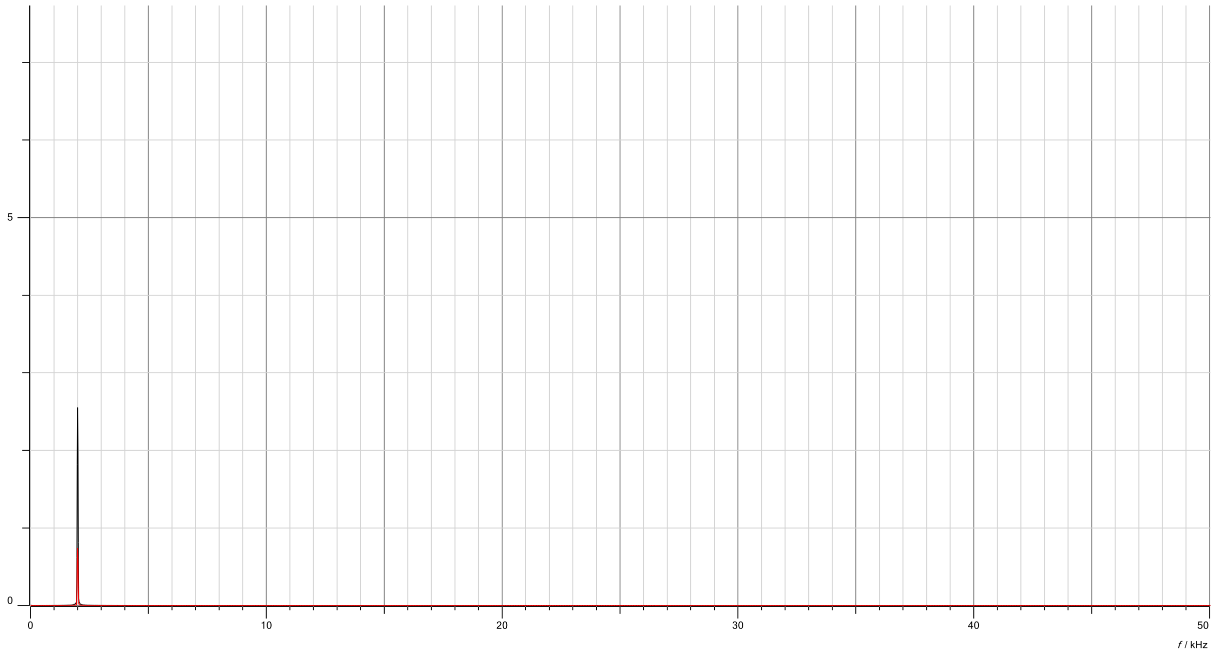

Figure 29: Non-Coherent Demodulation with filter in freq. domain

- When change the carrier phase from the phase shifter (φ) :

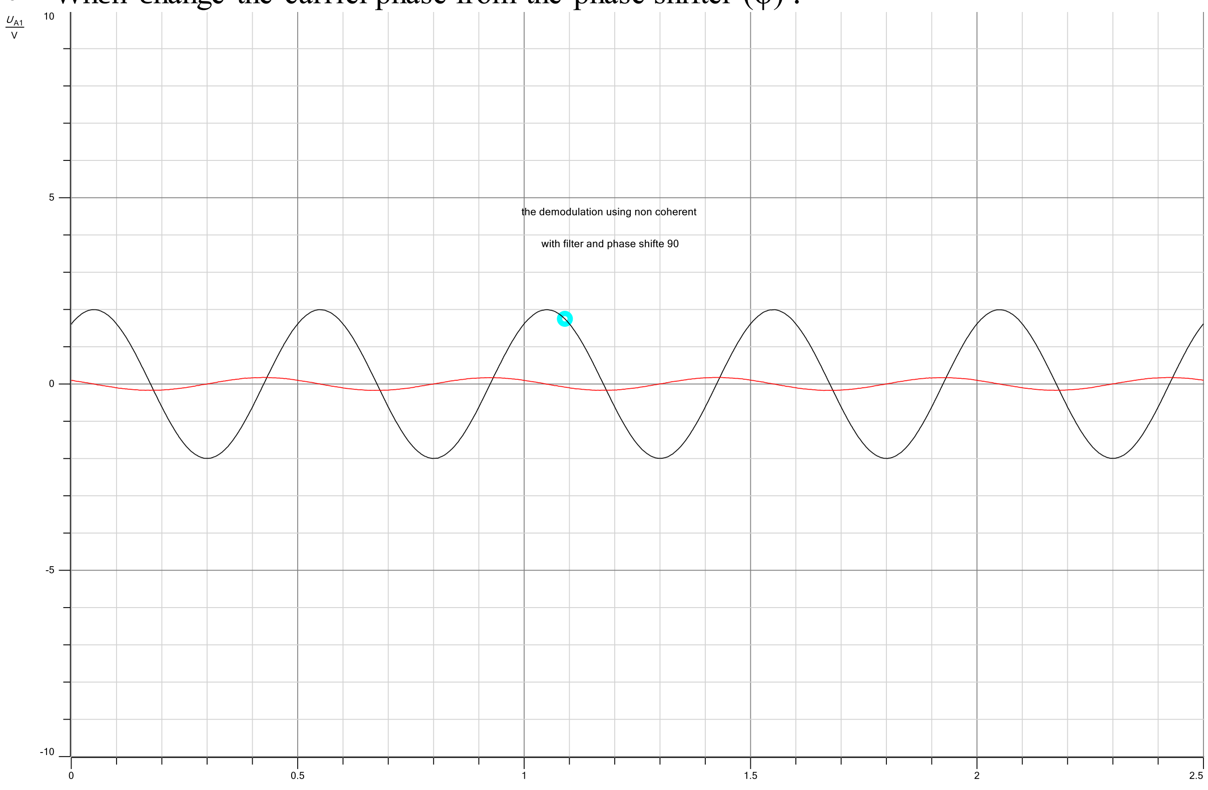

Figure 30: Non-Coherent Demodulation when change the phase

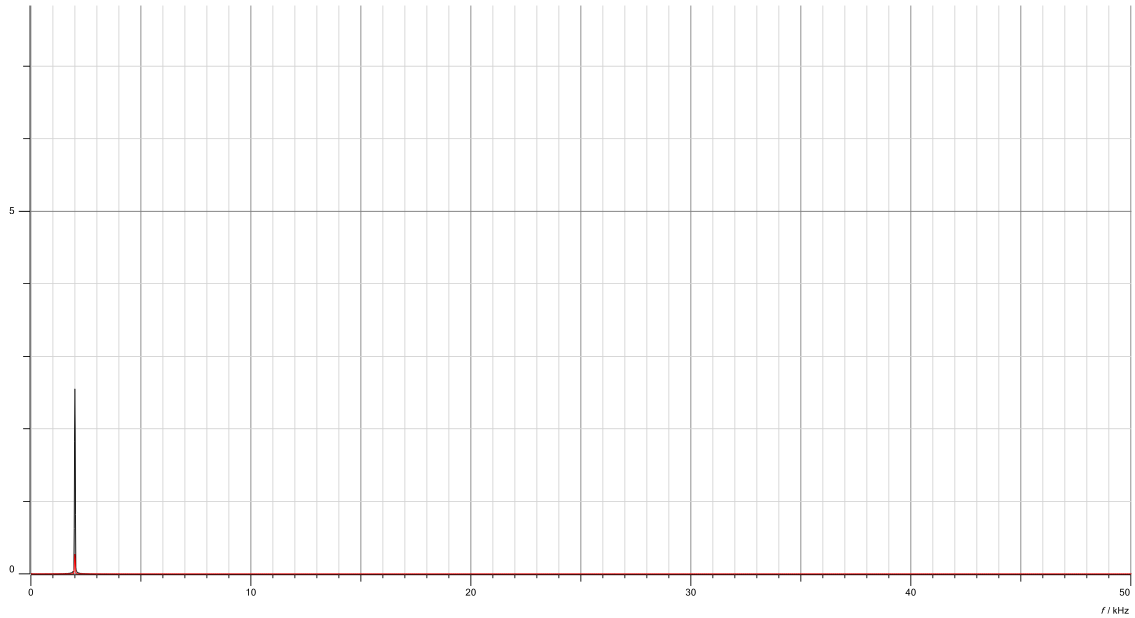

Figure 31: Non-Coherent Demodulation when change the phase in freq. domain

Except for a 90-degree phase shift, where the output becomes zero, when the phase of the signal changes, the amplitude of the rectified signal stays constant. In quadrature detection, this ability is utilized to demodulate signals without the necessity for coherent demodulation.

2.2 Single-side band Residual carrier

2.2.1 Modulation

The components have constructed and used as shown in the figure below:

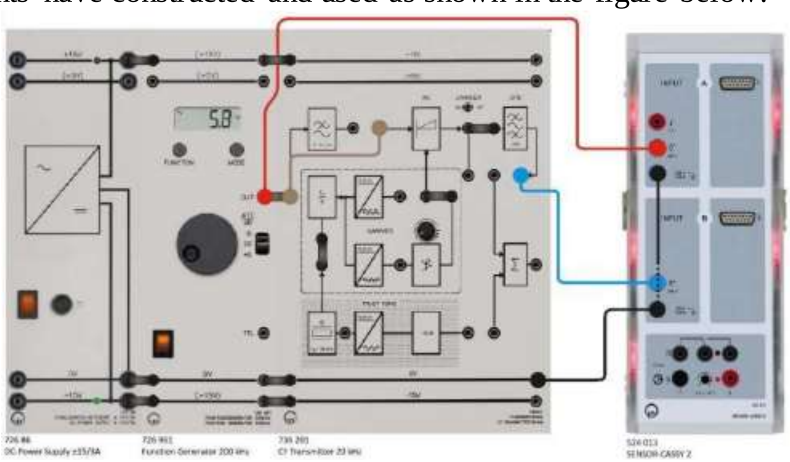

Figure 32: SSB modulation components

sine, VSS = 4V and fm = 2 kHz , and the toggle switch set to CARRIER ON because the carrier signal is needed at the receiver to demodulate the received signal and recover the original message signal.

Below is what occurred:

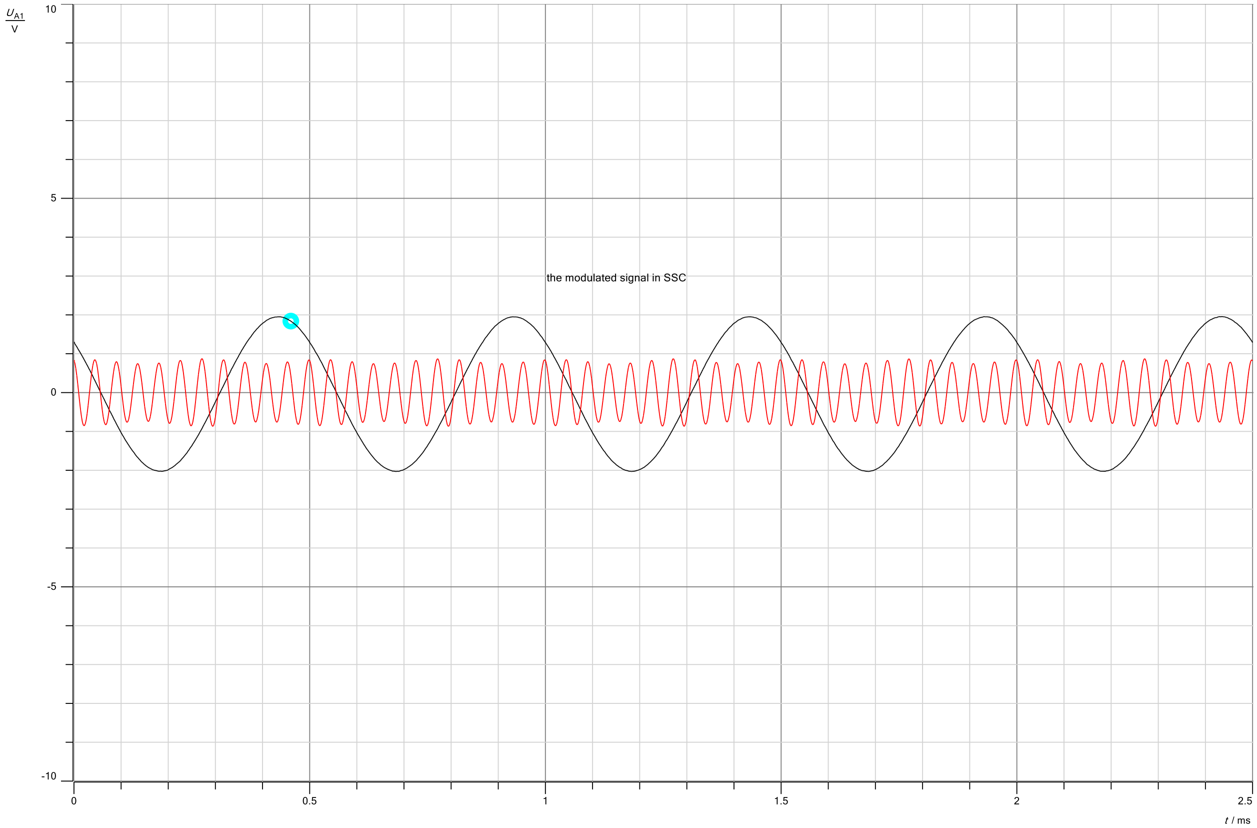

Figure 33: SSB output at 4v & 2kHz

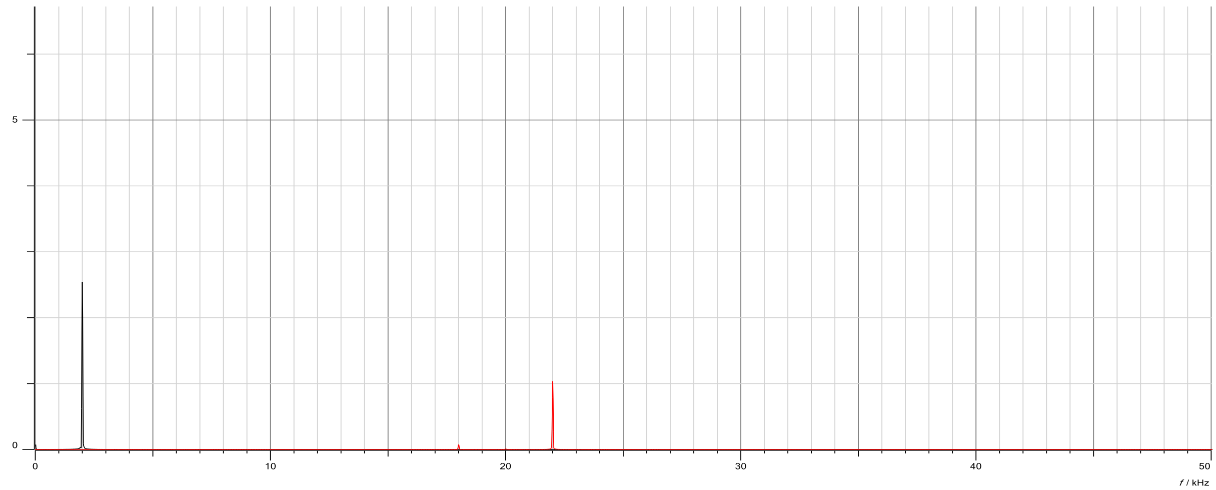

Figure 34: SSB output at 4v & 2kHz in freq. domain

The modulated signal is a combination of the message signal and one of its sidebands, with the carrier signal suppressed. It contains all the information from the original message signal but with reduced bandwidth.

2.2.2 Demodulation

2.2.2.1 Coherent Demodulation

The function generator set to: sine, VSS = 4V and fm = 2 kHz, and the components assembled as shown below:

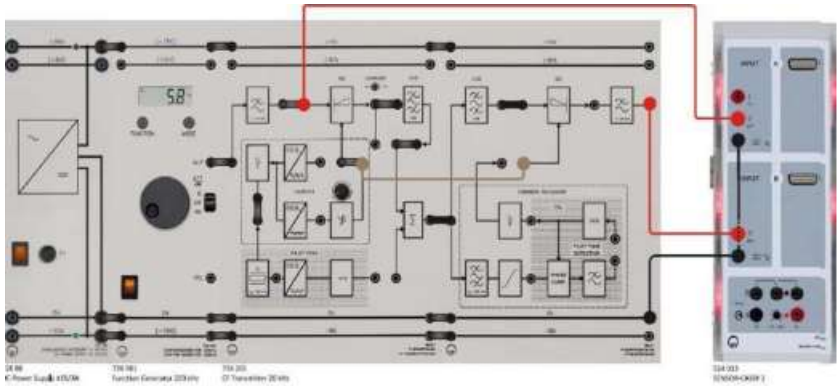

Figure 35: SSB Coherent Demodulation

Then the phase controller (φ)set to zero, and s(t) signal been fed directly into the demodulator. Using a wire feed a sinusoidal carrier signal into the auxiliary carrier input of the demodulator. Because a reference signal that matches the carrier frequency and phase of the transmitted signal is required. This enables the demodulator to retrieve the original message signal even if the carrier signal was suppressed during transmission.

Then we get this output:

- Before the output filter

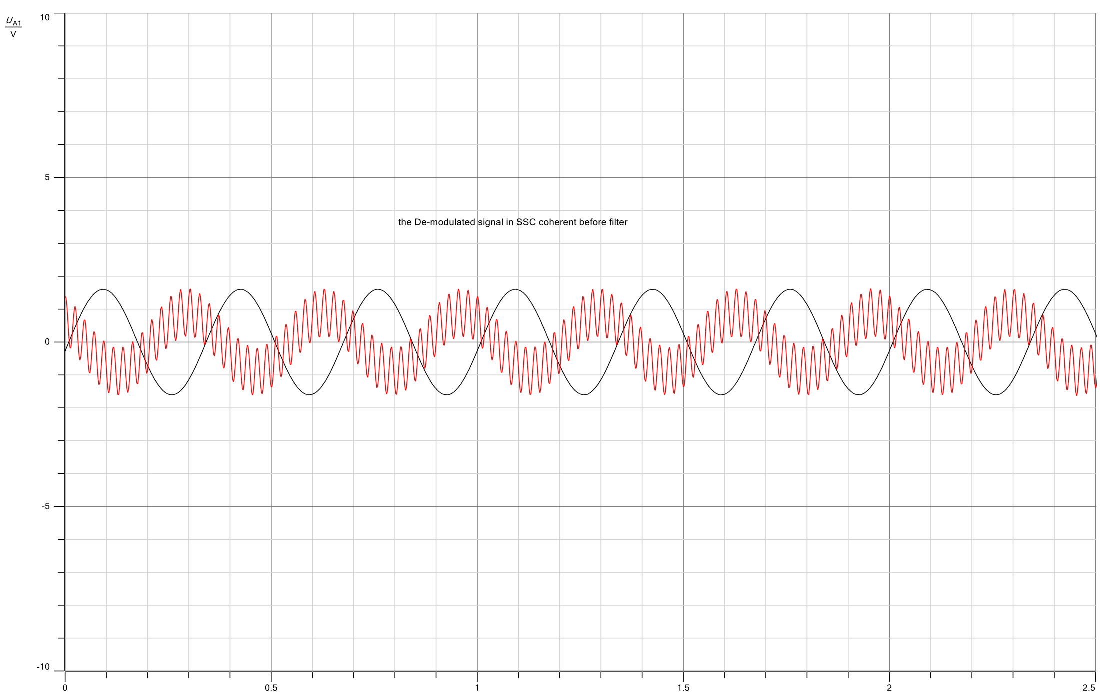

Figure 36: SSB coherent demodulation without filter

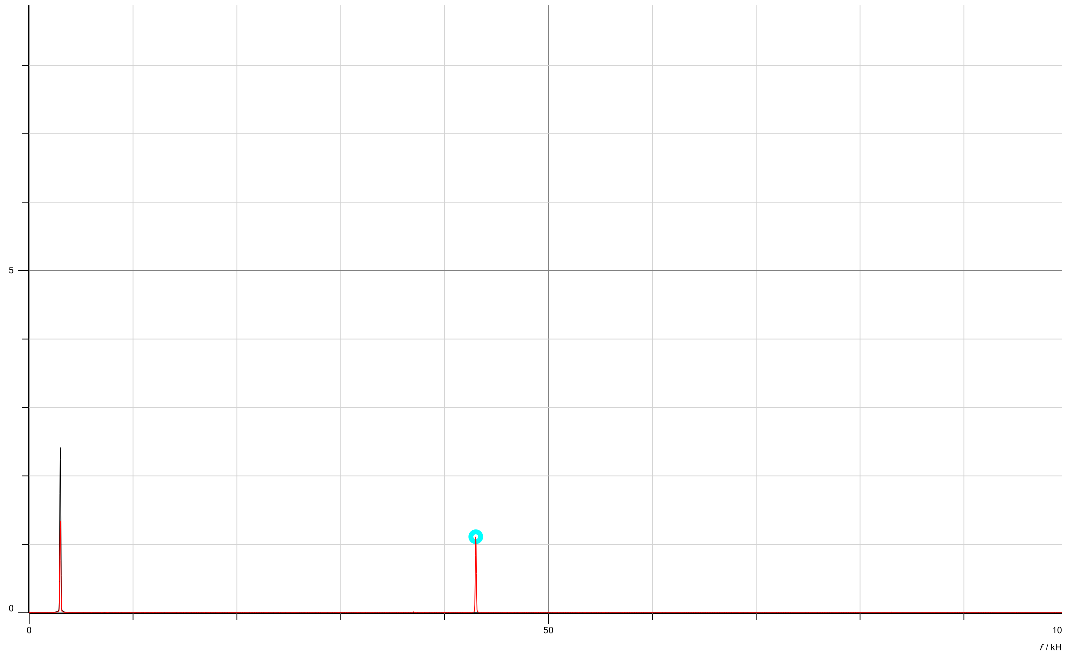

Figure 37:SSB coherent demodulation without filter in freq. domain

- After the output filter

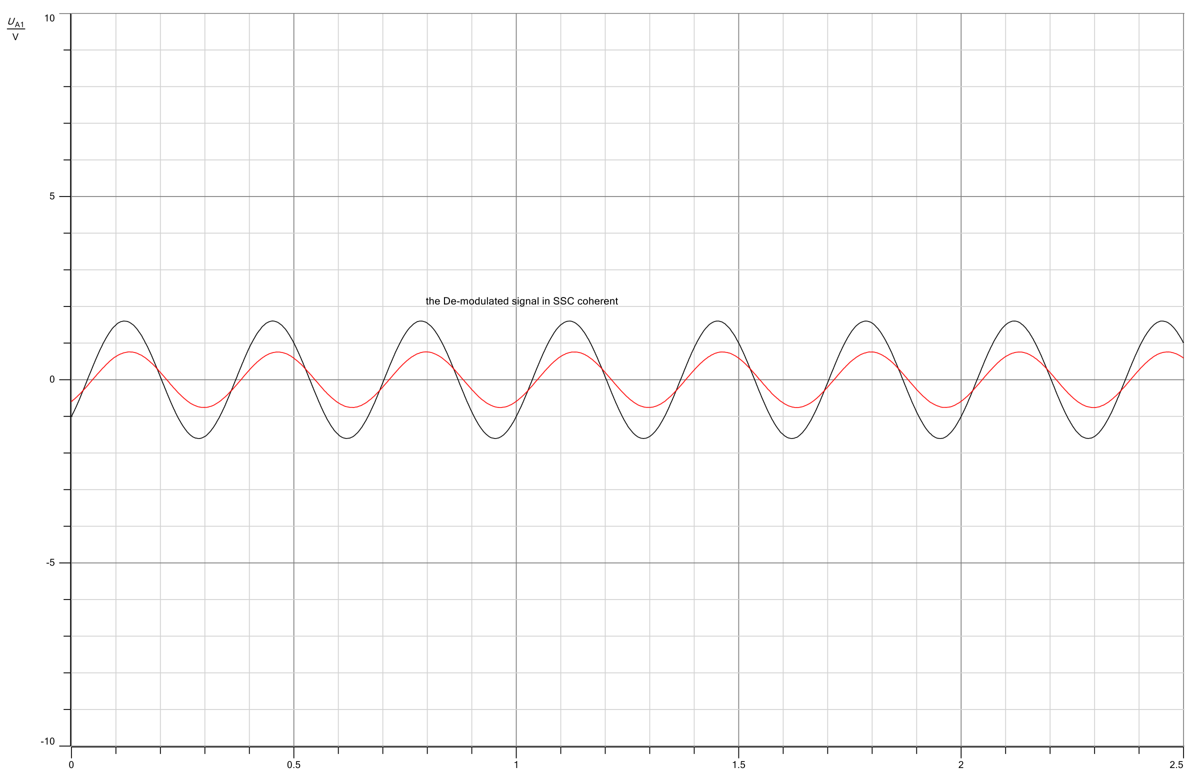

Figure 38: SSB coherent demodulation with filter

Figure 39: SSB coherent demodulation with filter in freq. domain

As seen in the previous pictures, in coherent demodulation of SSB transmissions, the receiver output filter is used to eliminate undesired frequency components and improve signal quality by sending the demodulated signal through a low-pass filter with a cutoff frequency lower than the carrier frequency.

2.2.2.2 Non-Coherent Demodulation

After making a few basic tweaks to the meshes of the components to obtain the non-coherent, we obtained the following:

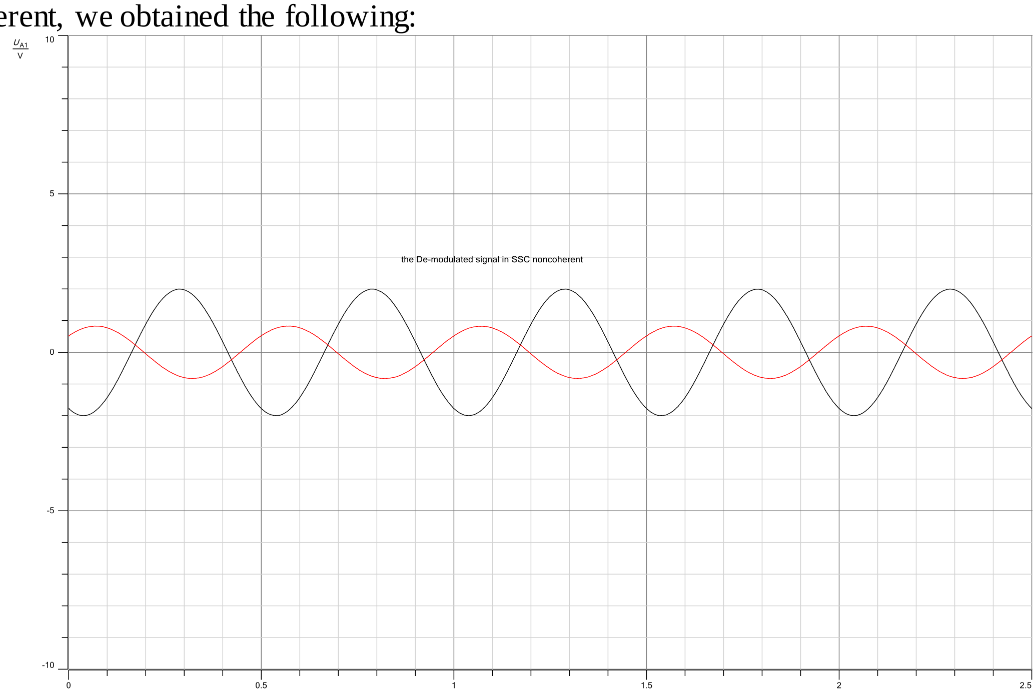

Figure 40: demodulated SSB non-coherent

Another phase shift :

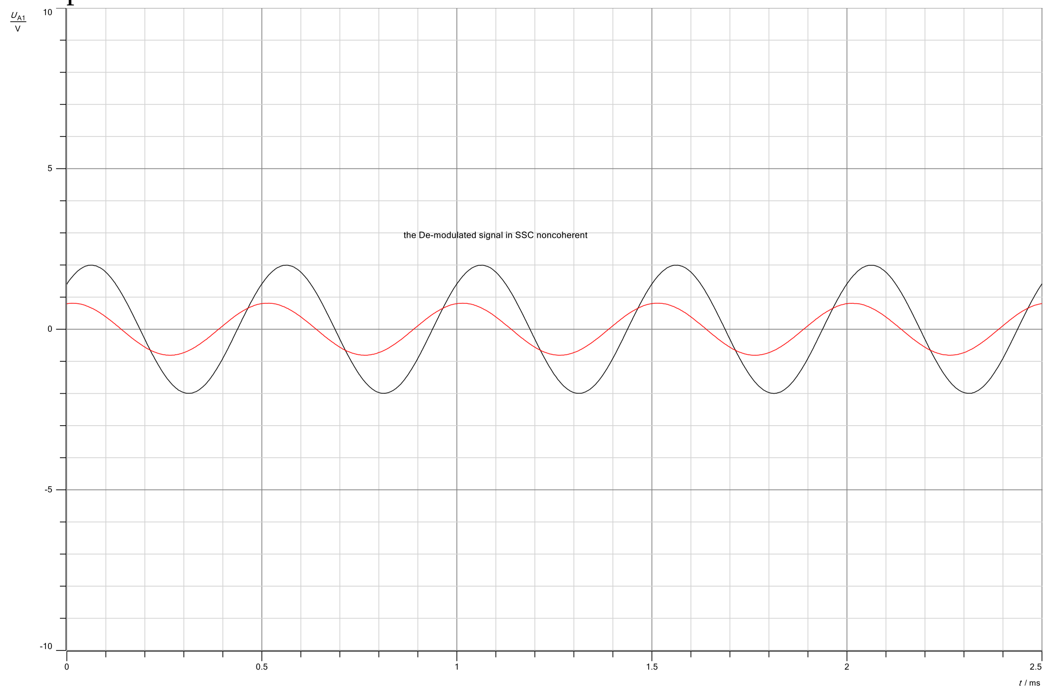

Figure 41: demodulated SSB non-coherent 2

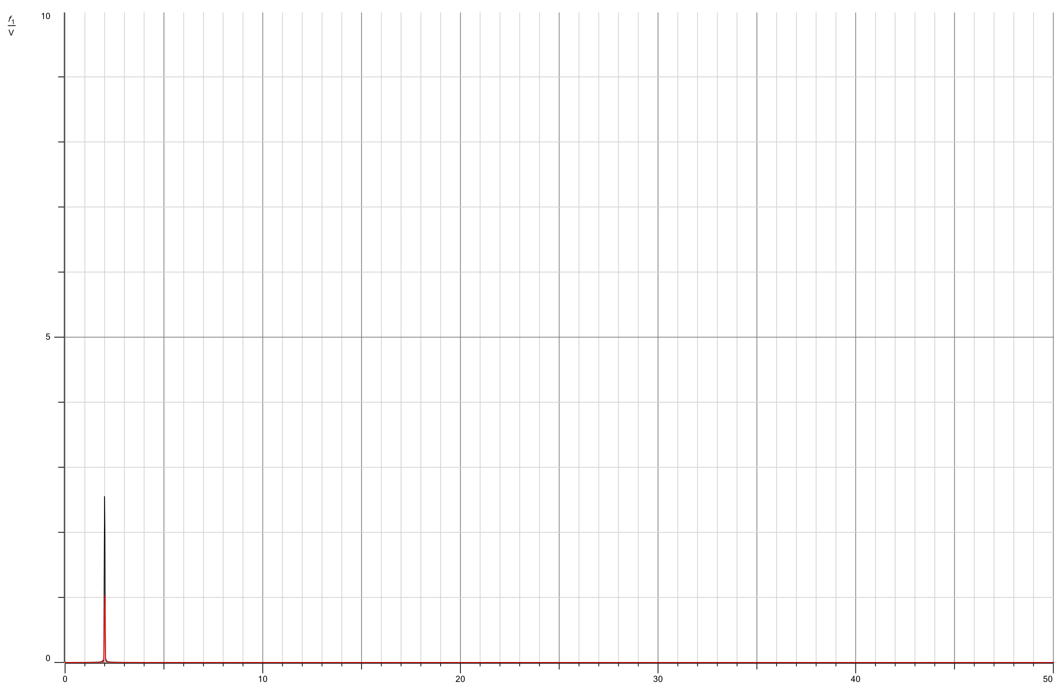

Figure 42: demodulated SSB non-coherent in freq. domain

We conclude from the pictures, that the phase shift between the received signal and the oscillator reflects the original message signal's frequency divergence from the carrier frequency. A 90-degree phase shift results in maximum output amplitude, whereas a 180-degree phase shift cancels the carrier signal, resulting in no output.

2.2.2.3 Phase Locked Loop (PLL) Coherent Demodulation

The Phase Locked Loop (PLL) is an alternative method for coherent demodulation where a receiver uses a loop to recreate a full carrier signal from an attenuated version of the original carrier. This regenerated carrier signal is then used for the demodulation process. The components were built and utilized as shows in the figure below:

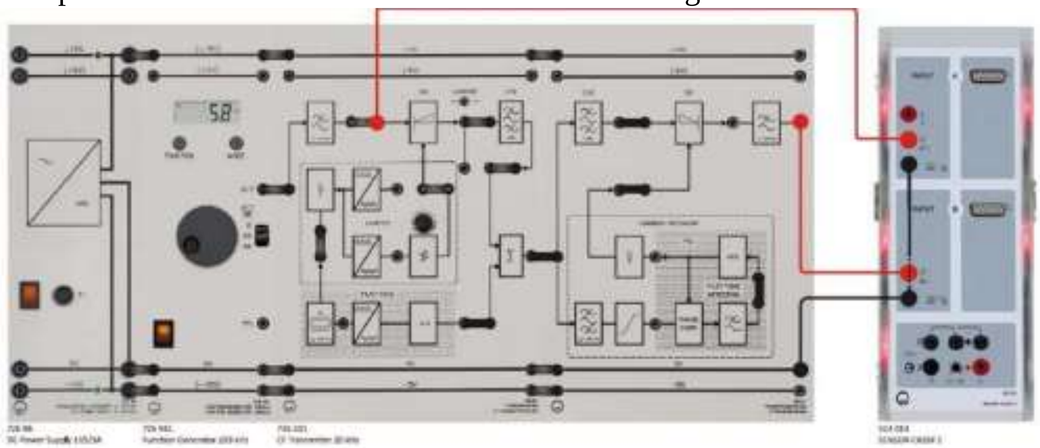

Figure 43: Phase Locked Loop (PLL) Coherent components

As a result, the following occurred:

- The pilot tone of the transmitter & the recovered signal in the receiver:

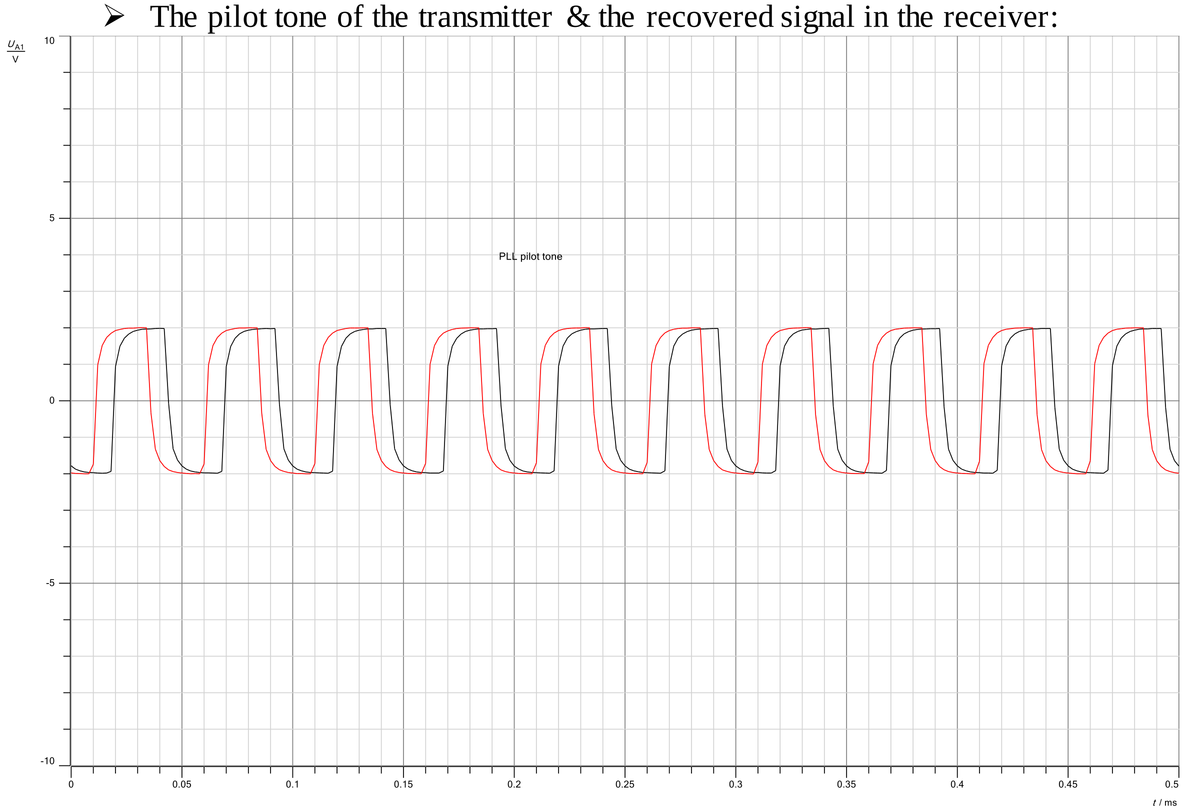

Figure 44: pilot tone - PLL coherent demodulation

That closely matches the original carrier signal, so that it can be used for coherent demodulation.

- Before the output filter:

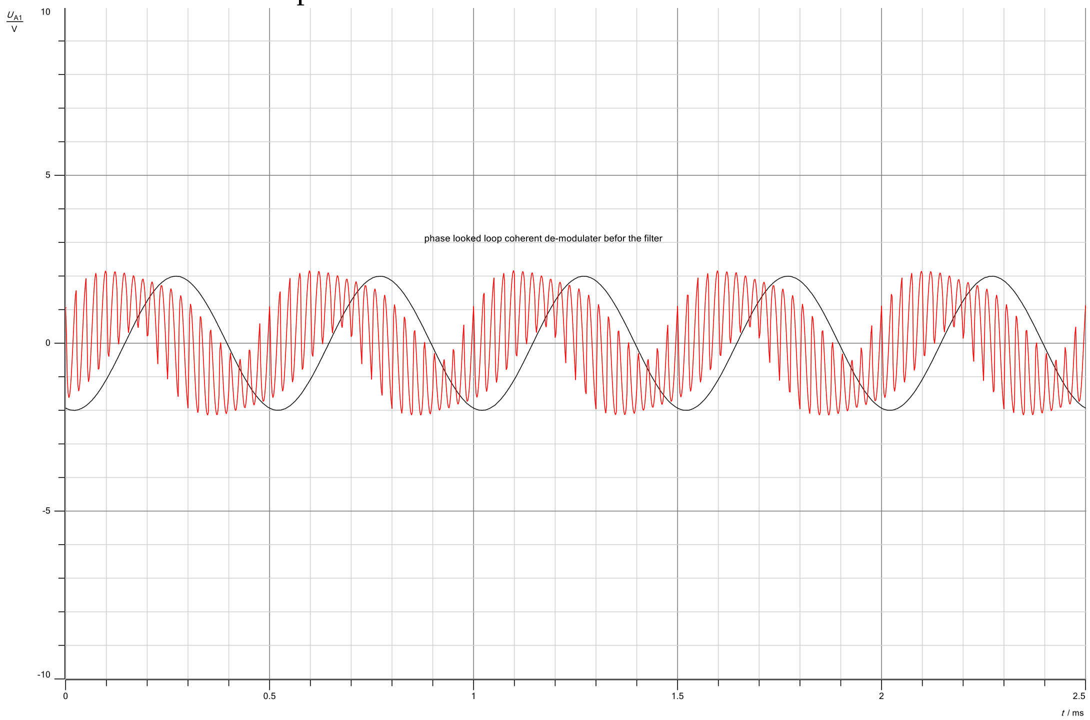

Figure 45: Phase Locked Loop (PLL) Coherent Demodulation without filter

- After the output filter:

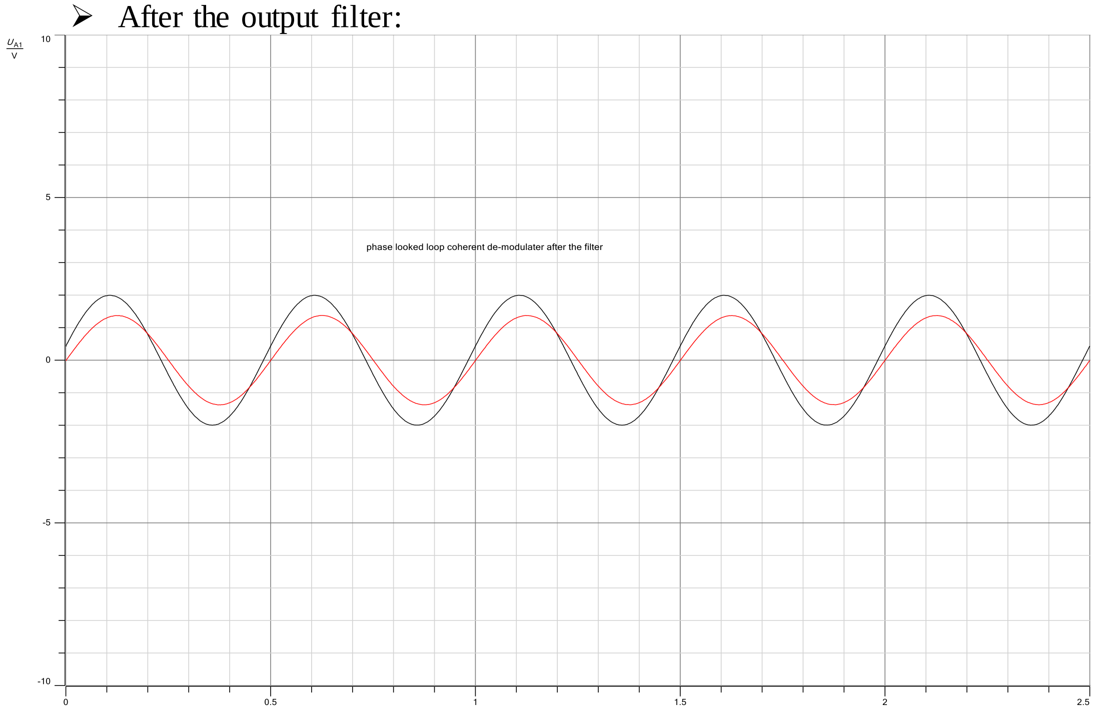

Figure 46: Phase Locked Loop (PLL) Coherent Demodulation with filter

So, the demodulation process using a Phase Locked Loop (PLL) can be successful .

3.Abstract

We tested Double Sideband (DSB) and Single Sideband (SSB) modulation approaches, as well as coherent and non-coherent demodulation techniques, as well as Phase Locked Loop (PLL) coherent demodulation in both time and frequency domains, in this experiment. We looked at how DSB and SSB approaches differed in terms of bandwidth utilization and efficiency, and additionally how changing the modulation index influenced the output signal. We employed envelope detection for non-coherent demodulation and examined the demodulated signal on an oscilloscope in both the time and frequency domains. PLL coherent demodulation performance was compared to various demodulation approaches, and the benefits and drawbacks of each method were examined.

4. References

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